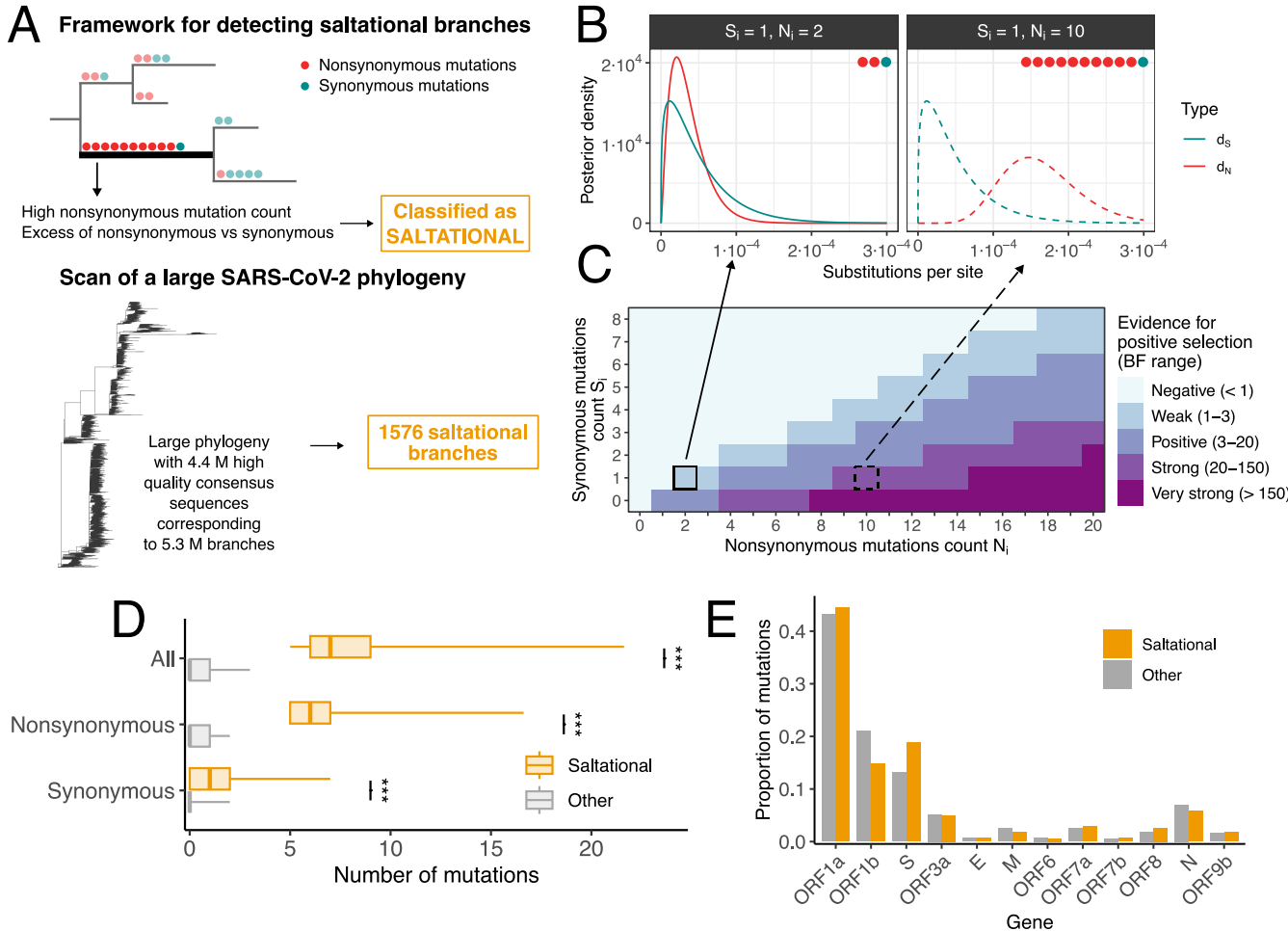

A
Framework for detecting saltational branches
Nonsynonymous mutations
Synonymous mutations
High nonsynonymous mutation count
Excess of nonsynonymous vs synonymous
Classified as SALTATIONAL
Scan of a large SARS-CoV-2 phylogeny
Large phylogeny with 4.4 M high quality consensus sequences corresponding to 5.3 M branches
1576 saltational branches
B
$S_i = 1, N_i = 2$
$S_i = 1, N_i = 10$
Posterior density
Substitutions per site
Type
d_S
d_N
C
Synonymous mutations count S_i
Nonsynonymous mutations count N_i
Evidence for positive selection (BF range)
Negative (< 1)
Weak (1–3)
Positive (3–20)
Strong (20–150)
Very strong (> 150)
D
All
Nonsynonymous
Synonymous
Saltational
Other
Number of mutations
E
Proportion of mutations
Saltational
Other
ORF1a
ORF1b
S
ORF3a
E
M
ORF6
ORF7a
ORF7b
ORF8
N
ORF9b
Gene